

ERA-NET: Aligning national/regional translational cancer research programmes and activities

## TRANSCAN-2

Joint Transnational Call for Proposals 2016 (JTC 2016) on:

**"Minimally and non-invasive methods for early detection and/or progression of cancer"**

# Pre-proposal Application Form

All fields must be filled in using Arial font, size 11, single-spaced.  
Applications should be submitted as a PDF file, formatted in DIN-A4.

**Please note that proposals either incomplete, using a different format or exceeding length limitations of any sections will be rejected without further review.**

**1a. Project title** (maximum 150 characters, including spaces):

Accurate validation of druggable biomarkers of Non-Small Cell Lung Cancer through liquid biopsy in early and locally advanced disease population.

**1b. Project acronym** (maximum 10 characters):

LIQUIDLUNG

**2. Project duration**(months):

36

**3. Project coordinator (research partner 1 in the consortium):**

|                                         |                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                                    | Christian Diego Rolfo                                                                                                                                                                                                                                                                                                                      |
| Country                                 | Belgium                                                                                                                                                                                                                                                                                                                                    |
| Position                                | Head of Phase I- Early Clinical Trials Unit - Antwerp University                                                                                                                                                                                                                                                                           |
| Institution/Department                  | Phase I Unit. Oncology. University Hospital of Antwerp. Belgium                                                                                                                                                                                                                                                                            |
| Address                                 | Wilrijkstraat 10, 2650 EDEGEM, Belgium                                                                                                                                                                                                                                                                                                     |
| Phone + Fax                             | +3238213646 – 38251592                                                                                                                                                                                                                                                                                                                     |
| E-mail address                          | <a href="mailto:Christian.rolfo@uza.be">Christian.rolfo@uza.be</a>                                                                                                                                                                                                                                                                         |
| Type of entity<br>(tick as appropriate) | <input checked="" type="checkbox"/> Academia (universities or other higher education or research institutions)<br><input type="checkbox"/> Clinical or Public Health (hospitals/public health and/or other health care settings and health organizations)<br><input type="checkbox"/> Small and Medium-sized enterprises (SME) or Industry |

## 4. Other research partners

| No. | Country | Name of research partner (principal investigator) | Institution, department & full address                                                                                                             | Phone & Fax                           | Email address              | Type of entity |                           |                 |
|-----|---------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------------------------|----------------|---------------------------|-----------------|
|     |         |                                                   |                                                                                                                                                    |                                       |                            | Academia       | Clinical or Public Health | SME or Industry |
| 2   | Spain   | M <sup>a</sup> José Serrano                       | University Hospital of Granada. Oncology Unit. Department: GENYO, Avda. de la Ilustración 114 18016 Granada, Spain                                 | +3495871550<br>0 Ext 123              | mjose.serrano@genyo.es     | X              |                           |                 |
| 3   | Italy   | Umberto Malapelle                                 | Department of Public Health, Molecular biology and cytopathology Unit. University Federico II of Naples Via Sergio Pansini 5, 80131, Naples, Italy | +3908174636<br>90                     | umberto.malapelle@unina.it | X              |                           |                 |
| 4   | Italy   | Diego Cortinovis                                  | Hospital San Gerardo Monza, Oncology department Via G. B. Pergolesi, 33, 20052 Monza MB, Italy                                                     | +39 039-<br>233.3683/039<br>-233.3203 | d.cortinovis@asst-monza.it |                | X                         |                 |
| 5   | Taiwan  | James Chih-Hsin Yang                              | National Taiwan University Hospital. No.1, Changde St., Zhongzheng Dist., Taipei City 10048, Taiwan (R.O.C.)                                       | +86<br>223123456<br>ext 67511         | chihsyang@ntu.edu.tw       | X              |                           |                 |
| 6   |         |                                                   |                                                                                                                                                    |                                       |                            |                |                           |                 |

5. Total requested funding: €994,724

## 6. Keywords

Please indicate three to seven keywords by using the [MeSH](#) vocabulary representing: the scientific content [(type of cancer; specific aim(s) and topic(s) (see Call Text, chapter 2. Aim of the call)]; the methodological and technological approach(es).

Non-small cell lung cancer, liquid biopsy, tumour markers, prognosis, mutations, allele frequency,

## 7. Project abstract (maximum 400 words)

Genetic profiling of solid tumours is currently undertaken using surgical or biopsy specimens. However, biopsies due to their invasive nature cannot always be performed routinely owing to their invasive nature and patient fragility. A single biopsy while providing a spatially and temporally limited snap-shot of a tumour can fail to reflect its heterogeneity, and lead to errors regarding patient treatment. A liquid biopsy, in this case blood sample, though, not only has the power to map the genetic landscape of all tumour lesions (primary and metastases), but very valuably opens up the systematic tracking of genomic evolution and with this, the earliest diagnosis of treatment resistance and the real time monitoring. Liquid Biopsy (LB) has now been verified with prognostic markers for disease progression in various cancer types, specially in NSCLC. The gold standard achievement of LB is 'Precision Medicine', where differences found among people of different racial, ethnic and familial backgrounds enable accurate identification of cancer risk, incidence and survival, and improved treatment decisions. LB overcomes a very significant challenge for clinicians treating lung cancer patients, the lack of tissue available for monitoring a patient over time.

This project has the clear aims of, primarily, (i) validation of the genetic druggable alterations present in circulating tumour cells (CTCs), RNA exosomal and free DNA isolates from patients' blood and (ii) validation of the prognostic value of genetic alterations detected on the same components for relapse. Secondly, looking at the correlation of the genetic profiles with the clinical-pathological characteristics observed, referencing standard biomarkers used in NSCLC and clinical outcome, to identify the best prognostic value.

The systematic approach presented here will validate the effect of the genetic alterations and the different allelic frequencies found among different races as novel biomarkers for predicting clinical outcome in NSCLC patients. With LBs being able to provide repeat 'low invasive' biopsies readily and tracking of patient genetic analyses over time with earlier assessment, this will improve patient risk assessment and adjustment of treatments. Outcome for health services will be also improved as the stratification of NSCLC patients improve clinical decisions and interventions. When these are adjusted to the individual risk of each cancer patient the patient outcomes will be improved their quality of life and life expectancies.

**8. Adherence of the proposal to the scope, aims and specific topics of the call** (see the Call Text, chapter 2. Aim of the call). Proposals will have to cover at least one of the specific areas listed under each undermentioned aim. Please, select as appropriate.

Aim 1. Risk stratification to distinguish groups by susceptibility for development or progression of cancer based on molecular biomarkers and established cancer risk factors:

- ☐ A. Risk stratification for cancer development  
☒ B. Risk stratification for cancer progression

Aim 2. Validation of multiparametric methods using the combination of promising biomarkers to improve our capability for early detection or progression of cancer:

- ☒ A. Molecular tumour markers  
☒ B. Imaging techniques  
☐ C. Bioinformatics techniques

Aim 3. Improve clinical evidence of minimally invasive methods:

- ☐ A. Analytical validity  
☒ B. Clinical validity  
☒ C. Clinical utility  
☐ D. Ethical, legal, and social implications (could also be considered)

Has the project been submitted elsewhere?

- ☐ Yes  
☒ No

## 9. Project description (maximum 3,000 words)

### a. Description of the project: medical need

Lung cancer (LC) remains a very significant cause of cancer-related deaths, with over 1 million deceased reported worldwide every year, the overall survival rate for patients with LC below 15%. Despite advances in treatment and diagnosis, it still has a very poor prognosis. It has been reported that around 60% of lung adenocarcinomas carry oncogenic driver mutations. The study of these mutations has provided new drug targets and novel drugs (1).

One of the main targets is the **Epidermal Growth Factor Receptor (EGFR)** and its signaling pathways, which are well known to be actively involved in NSCLC disease progression through angiogenesis, cell survival, proliferation and metastasis. EGFR mutations are the most frequent driver mutation in NSCLC; it has been reported to be present in 40-60% Asian, 15-20% Caucasian and around 30% Latin-American patients. This discovery has led the development of targeted therapies against the EGFR mutated form, the tyrosine kinase inhibitors (TKIs) such as gefitinib, erlotinib or afatinib (1).

Other mutations have now been identified as valuable in targeted therapies in NSCLC. One of these is the **ELM4-ALK** fusion gene, detected in 3-8% of NSCLC patients (1). This mutation leads to the synthesis of a fusion protein which induces a constitutive activation of the ALK receptor and consequently its downstream pathway. The discovery of this translocation led to the development of the ALK inhibitor crizotinib, approved for use in advanced NSCLC ALK+ patients. Other potential targets for directed therapy include point mutations in **BRAF**, **K-RAS**, **ROS1**, **PIK3CA** or **NTRK1** genes.

The mechanisms of resistance, which strongly decrease the effectiveness of the TKI inhibitor treatment, worsen patient prognosis (2). The detection of oncogenic driver mutations is mostly identified through tumour tissue based tests, using techniques such as immunohistochemistry (IHC), fluorescent hybridization in situ (FISH) and more recently, next generation sequencing (NGS). These techniques, although useful, still depend on there being sufficient tumour tissue recovered in the biopsy, where a tumour biopsy has been performed. Moreover, the tumour biopsy taken cannot represent the whole molecular profiling of the tumour. Targeted therapies in NSCLC will greatly benefit such patients, if only earlier diagnosis and more sensitive detections of oncogenic driver mutations could be detected, and then monitored over time as the treatment progresses. This would allow to anticipate clinical and radiological progression detection improving the treatment choice.

It has been recognised now for some years that the problems noted above could be potentially solved thanks to testing Liquid Biopsies' (LB). In conjunction with new detection technologies, clinically relevant information from these **(LB)** has created enormous interest (3). The LB concept promises to reveal clinically relevant information, in a non-invasive fashion, analysing blood, saliva or other body fluids of cancer patients.

Tested so far in liquid biopsies are circulating tumour cells (CTCs), circulating-tumour DNA (ctDNA) and exosomes (for tumour exosomal nucleic acid). They can be a reliable and trustworthy source of information about tumour heterogeneity, treatment response and early stage detection.

**CTCs**(3) Circulating tumour cells (CTCs) are cells that have separated from the cancer primary site or metastatic, and disseminate into the peripheral blood circulation. CTCs are understood to be part of the long process of cancer metastasis. It is well established that LC releases only a limited number of CTCs. This makes their analysis difficult, due to poor methods of isolation. However, thanks to the improvement in knowledge about CTCs, detection and isolation methods. In general, the value of studies of CTCs in solid tumours include (a) risk prediction for metastatic progression and relapse, (b) staging determination of success and real-time observation of responses to therapies, (c) determination of therapeutic targets and mechanisms of resistance and (d) understanding and tracking the metastasis process in solid tumour patients.

**The ctDNA**(3) is released into the blood stream as a result of apoptotic or necrotic processes in the tumoral cells. This is increased in cancer patients due to both a higher replication rate of the tumour cells and the ineffective removal of apoptotic and tumour cells, resulting in higher levels of ctDNA in the peripheral blood. Thanks to new analysis techniques such as digital droplet PCR (ddPCR) and NGS, it is now possible to detect molecular alterations with clinical value. In 2016, the FDA approved a companion diagnostic blood test to detect the EGFR mutation in ctDNA for treatment of NSCLC patients with erlotinib. Other alterations have been described in ctDNA such as, KRAS (1).

**The exosomes** (3) Exosomes are a class of extracellular vesicles released by all cells, with a size range of 40-150 nm and a lipid bilayer membrane. Exosomes contain DNA, RNA, and proteins. Exosomes are found in significant numbers in body fluids including the blood. Additionally, they may mediate specific cell-to-cell communication and activate signalling pathways in cells they fuse or interact with. Exosomes are detected in the tumor microenvironment, and emerging evidence suggests that they play a role in facilitating tumorigenesis by regulating angiogenesis, immunity, and metastasis.

CTCs, circulating ctDNA and exosomes (LB) could identify their risk for metastatic relapse or metastatic progression. Moreover, these components in NSCLC patients could display a set of genetic and epigenetic differences common for tumours including oncogene activation, chromosome loss, and tumour-suppressor gene silencing by methylation.

#### **b. Description of the project aims**

1. To analyse the targetable genetic alterations (L858R, T790M in EGFR; V600 in BRAF; and EML4-ALK fusion gene status) in liquid biopsy (CTCs, ctDNA and exosomes), using the most innovative molecular and genetic technologies available today, in patients with advanced NSCLC. The main objective is to validate the presence of these genetic alterations in LB as prognostic markers for stratification of risk of relapse in patients with NSCLC.

2. To validate the known cancer biomarkers (L858R, T790M in EGFR; V600 in BRAF; and EML4-ALK fusion gene status) among Caucasian, Oriental and Hispanic populations. This will identify genetic diversity and improve the 'precision medicine' that can be practised by clinicians. It is known that allele frequency, is different among diverse populations. The project aims to validate selected markers among different populations and to identify within this group those universal markers in cancer diagnosis.

To achieve our objectives, we will perform the following activities:

- CTC isolation and genetic profiling of EGFR, BRAF, KRAS, ROS1, PIK3 and NRTK1 genes via NGS using Illumina NextSeq 500.
- Exosomal RNA isolation and transcriptome analysis, performing gene transcription analysis. Detection of fusion gene: ELM4-ALK
- ctDNA isolation and target mutations detection of EGFR (exon 19, exon 21), BRAF, KRAS, ROS1, PIK3 and NRTK1 genes using ddPCR (QuantStudio™ 3D Digital PCR System):
- Analysis of the data and their correlation between primary tissue and liquid biopsy.
- Bioinformatic validation for a universal non-invasive marker for lung cancer monitoring.

**c. Statement of the research hypothesis**

Circulating biomarkers, such as circulating tumour DNA (ctDNA) circulating tumour cells (CTCs) and exosomes are expected to significantly enhance our knowledge about the metastatic process in NSCLC. Differences in cancer risk, incidence and survival among people occur due to different racial and ethnic. The detection of the allelic frequencies in different populations, will enhance health services abilities to predict not only the cancer risk more accurately but also the evolution of disease (5). Better and earlier risk stratification of patients will produce improved patient outcomes in quality of life and life expectancies.

Additionally, the improved genetic analysis based on liquid biopsies and analysing multiparametric clinically valuable biomarkers will improve treatment options by focusing on the very cells that are the direct precursors of distant metastatic disease, and inherently different than the primary tumour's biology.

To shift the current clinical paradigm assessing tumour biology in real time by molecular profiling of LB will serve to predict treatment response, monitor patients for emergence of drug and treatment resistances, and may even assist in the identification of novel therapeutic targets.

**d. Preliminary data:**

Regarding the study of CTCs, a number of studies have concluded that the baseline CTCs count is an independent negative prognostic factor, and correlates, shorter progression-free survival and overall survival. In advanced stage cancer (IIB-IIIB) the presence of CTCs has been correlated with an increased risk of early recurrence (4). In a recent study comparing tumour biopsies with simultaneously collected CTCs and ctDNA, the resistance-associated mutation T970M was detected in 47% to 50% of patients respectively, with a concordance ranging from 57% to 74% (2). In those patients where the paired biopsy was negative or inconclusive. All ALK-positive patients were found to have four or more ALK-rearranged CTCs. Similarly, ROS1 rearrangements, another target for crizotinib-based therapy, have been detected in the CTCs, with positive paired tumours for ROS1-gene (1). Furthermore, the circulating levels of tumour exosomes, exosomal small RNA have been tested so as to validate them as potential biomarkers for both diagnosis and prognosis in adenocarcinoma patients. (1).

**e. Methods****Experimental design**

Samples will be collected in the five institutions, University Hospital Virgen de las Nieves, University Hospital of Naples, University Hospital of Antwerp, San Gerardo of Monza, Hospital and National Cancer Research Center Taiwan University after the approval by their Ethical Committees.

The total number of patients will be 100. The target population will include patients, stage IIB, IIIA and IIIB, both treated and untreated. The patients included will be from different races to find differences among races. The total amount of blood per patient needed for the project will be 30ml. A guideline for the correct blood extraction will be sent to the centres together with a clinical data sheet to fill at time of extraction.

**Inclusion criteria**

- ☐ Histologically or cytologically confirmed diagnosis of NSCLC stage IIB, IIIA, IIIB
- ☐ Age 18 years or older at the time informed consent given.
- ☐ Patients may have received or not received one prior treatment.
- ☐ Patients have a World Health Organization (WHO) performance status 0-2.
- ☐ Patients have the ability to understand and provide signed informed consent.

**Exclusion criteria**

- ☐ Presence or history of a malignant disease other than a NSCLC.
- ☐ Stage IV
- ☐ Patient does not have the ability to understand and provide signed informed consent.

**Biopsy analysis:**

Next Generation Sequencing.

The analysis of the tissue will be performed in multiple in the National Taiwan University under the supervision of Prof. James Yang.

**1) CTCs analysis**

10 ml extraction of blood will be stored in STRECK specific tubes. ISOFLUX® technology (FLUXION) for isolation will be used to isolate CTCs, according to the protocol established by the group of Liquid Biopsy and Metastasis of the GENYO centre (4), and with the additional recommendations of the ISOFLUX equipment supplier. This protocol will be established for the detection and genetic characterization via NGS of the targeted genes (see above) of circulating tumour cells.

**2) cfDNA analysis**

10 ml of blood samples will be stored in Citrate specific tubes. The blood sample will be processed using one step of centrifugation to separation of plasma. Then, the plasma will be stored at -80°C until use. The extraction of cfDNA will be performed using DNeasy blood kit (Qiagen)/ QIAamp Circulating Nucleic Acid kit (QIAGEN) according to the manufacturer's instructions. DNA will be quantified with the Qubit 2.0 Fluorometer and Qubit dsDNA HS Assay kit (Life Technologies, Carlsbad, CA) according to the recommended protocol. Finally, cfDNA target mutations detected by IonChef Sequencing and 316 PGM chip v 2 (ThermoFisher).

**3) Exosomes analysis**

10ml of blood will be stored in EDTA tubes. Isolation of exosomal RNA from plasma through commercial kit (ExoRNeasy mini kit, Qiagen©) and quantization by Nanodrop® (Thermo Scientific©). The RNA collected will be used for the NGS analysis of the mutations and the EML4-ALK transcription. For isolation of exosomes a commercial kit (ExoEasy kit, Qiagen©) will be used.

**4) Bioinformatics and statistical analysis of all results. (Bioinformatic unit. GENYO Centre)**

Workflow for SNP calling from NGS data will cover QC (using fastqc application) alignment of reads with the reference genome with the BWA algorithm, snp calling with samtools, variant annotation and filtering with the annovar application, for an overview of algorithms see (Pabinger et al., A survey of tools for variant analysis of next-generation genome sequencing data, Brief Bioinform 2014) will be used to detect somatic mutations in the gene panel. Regression models will be applied to evaluate the association of the discovered variants with the clinical variables.

The main steps of the rna-SEQ workflow will cover QC (fastqc), read alignment (RSEM) and DESeq2 or EdgeR for differential expression analysis

#### **f) Novelty and originality of the project**

Our proposed LiquidLung will constitute the first study that will compare and validate the prognostic value of CTCs/cfDNA and exosomes in an multicentric work to identify the relevance of the “liquid biopsy” to personalized cancer medicine. While there are other studies where these biomarkers have been analysed individually, this study will be the first where the three are done simultaneously, hence increasing the power of the analysis. On top of that, this study will be able to identify different allelic frequencies found among different races, a key element to predict clinical outcome in NSCLC patients.

#### **g) Feasibility of the project:**

The present proposal is based on the clinical and translational experience that each of the partners have in the analysis of the three different types of LB demonstrated in the publication of the members of the consortium.

All partners will participate in WP1 (Project management) and WP2 (Patient recruitment and sample collection) which will be coordinate by the leader of the Project Coordinator, Prof. Dr. Cristian Rolfo. His current role as Director of Clinical Trails Management Program at the institution offer certainty to ensure the correct management of the project and the correct handling of the patient samples. The rest of partners have also extensive experience in handling patient samples and all of them will present to their respective ethic committee the project in order to be approved before sample collection start.

The molecular analysis of the blood samples (WP3) will be done in three different countries. CTC analysis will be done by the Liquid Biopsy Group led by Dr. Serrano at GENyO Center in Granada, Spain. Her group has a large experience in isolating and characterizing phenotypically and genotypically CTCs. They also have access to all platforms needed to do so (Fig 1)

cfDNA will be analyzed by the Italian group led by Prof. Malapelle. They have worked extensively in circulating biomarker, tailoring NGS assays (i.e., EGFR mutations, ALK traslocation, PDL – 1 expression) for a number of different applications using both solid tissue and body fluids as DNA sources.

The group of Prof. Rolfo will be in charge of the exosomal analysis of the samples. Exosome characterization is relatively new and the group of Prof. Rolfo is one of the few European groups which have a track record working with them.

The group of Prof. Yang will perform the NGS analysis in tissue. The experience of this group of Taiwan Cancer Center is demonstrated in the biography of Prof. Yang.

With the fusion of the clinical and basic experience in this field we will be able to obtain a real information of the clinical value of each of these tests. The expected results of the research are transferable to the organization, resource management, health services or health policies and they could be transferable through consensus documents, published clinical practice guides and applicable in the Health Systems.

#### **h. Work Plan (Fig. 2)**

##### **Work Package 1 (WP1) Project Management**

###### **Objectives**

- To guarantee the scientific, non-scientific and administrative coordination of all activities of the Project.
- To ensure the timely realization of all project objectives, milestones and deliverables



- To ensure the achievement of appropriate quality standards
- To ensure that the EU legal contractual obligations are met in accordance with the grant agreement

**Deliverables (Months)**

- Project Handbook (M1)
- Periodic Report (M6/M12/M18/M24/M30)
- Final Project Report (M36)

**Work Package 2 (WP2) Patient recruitment and sample collection****Objectives**

- Patient selection
- Informed consent signatures from patients
- Patient accrual and clinical data form compilation
- Sample extraction and preparation

**Deliverables (Months)**

- Database with Clinical information of patients (M1)
- Updating the database (M6/M12/M20)

**Work Package 3 (WP3) Molecular analysis****Objectives**

- Molecular characterization through NGS of Circulating Tumor Cells
- Molecular characterization of cell-free DNA
- Molecular characterization of exosomes from plasma

**Deliverables (Months)**

- Planning samples (M1)
- Database with Clinical and biological information of the patients (M1/M12/M24/M34)
- Standard working protocols (M1)

**Work Package 4 (WP4) Data analysis**

**Objectives**

- All the data obtained from the experiments will be set in common to better understand not only the individual power of each compartment but also to understand the liquid biopsy as a whole.
- Statistical analysis of the results.

**Deliverables (Months)**

- Data generated from bioinformatics and statistical analysis will be submitted in internal report for all the institutions (M12/M24/M36).

**i)Potential impact of project**

Druggable targets have a predictive and prognostic value in NSCLC subpopulations when diagnosed in tissue biopsies. Our results will open an important window of opportunity in these patients, allowing the possibility to get this information in a non-invasive way. The results of our research will represent a step forward for a clinical application of all the liquid biopsies family members. The identification of these biomarkers in early and locally advanced disease will have an impact in the illness prognosis. The validation of LB methods will have a worldwide impact for the patients and scientific community.

**j)References**

1. Ravi Salgia. Mutation testing for directing upfront targeted therapy and post-progression combination therapy strategies in lung adenocarcinoma. Expert Review of Molecular Diagnostics. 2016. 16, ( 7), 737-49.
2. Jessica J. Lin, et al. Resisting Resistance: Targeted Therapies in Lung Cancer. Trends in Cancer. 2016; 2 (7), p350–64.
3. Domenico Trombetta et al. Liquid biopsy and NSCLC. Future Medicine. 2016, 5, (2) ,91-104.
4. Bayarri-Lara C, et al. Circulating Tumor Cells Identify Early Recurrence in Patients with Non-Small Cell Lung Cancer Undergoing Radical Resection. PLoS ONE.2016;11(2): e0148659.
5. Henderson BE,etal. The influence of race and ethnicity on the biology of cancer. Nature reviews Cancer. 2012;12(9):648-53.

**10.Capacity building activities (if eligible for the funding organization / country) (maximum 300 words)**

Please specify whether the project will include capacity building activities. If so, please describe the nature and purpose of the planned activities taking into account information described in section 2.2 of the Call Text. The budget will have to be mentioned in the financial plan (sections 12 and 13) in the appropriate line.

**11. Brief CV for each partner in the research consortium** (i.e. the project coordinator and each principal investigator) including a description of the main domain of research and a list of the five most relevant publications within the last five years regarding the proposal (once converted into PDF document: max 1 page for each partner).

**Partner 1. Prof. Christian Rolfo**

**Professor in Oncology Antwerp University Hospital , Belgium**

He was born in Córdoba, Argentina in 1972; studied at the *National University of Córdoba*, graduating from medical school in 1996. In 1998 he began his studies at the *University of Milan*, Italy. He got a scholarship for a project in Breast Cancer with the prestigious Dr. Luca Gianni, in the National Cancer Institute of Milan. He trained with Drs. Lisa Licitra and Paolo Casali. During this period, he dealt with caring and Phase I Research Protocols, under the direction of Prof. Gianni. Obtained the European Oncology Board certification in 2003 with a thesis based on a Phase I trial with the combination of a new taxane and doxorubicin in Breast Cancer. Dr. Rolfo is also a Doctor of Medicine and Surgery at the *University of Catania, Italy*, with a thesis based on a Phase I trial and pharmacokinetics of BMS 184476 and anthracyclines in solid tumors. In 2004 he worked at the Oncology Unit of the Clinica Rotger until September 2012 under the direction of Prof. Rafael Rosell in Lung cancer. Actively involved in studies of molecular biology and clinical research in lung cancer, including liquid biopsy. In 2009 obtained the PhD degree and Doctor Europaeus in Clinical and Experimental Oncology Research with *cum laude* from the *University of Palermo*, Italy, for the thesis: Relationship Between metabolic activity (SUV max) by 18-FDG PET and Histology, stage, EGFR mutations and survival in Patients with NSCLC, under the direction of Prof. Antonio Russo and Prof. Rafael Rosell.

From October 2012 is Senior Staff Member in the Department of Oncology as Professor at the University Hospital Antwerp, Antwerp University in Belgium, headed by Prof. Marc Peeters. Currently he is Head of Phase I - Early Clinical Trials Unit, Director of Clinical Trials Management Program as well. He is focus on Clinical Research, Drug Development and Resistance, including new biomarkers research (immune-related biomarker as CD70 and mutations as TP53, EGFR and ALK and cMET alterations.). Additionally, he is actively working in Liquid Biopsies in Lung Cancer and pancreatic cancer, specifically is leading an exosomes research program at the University of Antwerp and is member of circulating tumor DNA working group.

He completed his training in organization on Phase I at Department of Investigational Cancer Therapeutics, Division of Cancer Medicine, The University of Texas MD Anderson Cancer Center, Houston, TX (Visiting Professor Program) with Prof. David Hong.

Prof. Rolfo also obtained the Master degree in Business Health Administration by Polytechnic University of Valencia, Spain in 2010. He was appointed in February 2011 as visiting professor in Medical Oncology by the Molecular and Clinical Genetic Oncology Unit at the Interdepartmental Center of Research in Clinical Oncology, School of Medicine, University of Palermo (Italy). Prof. Rolfo is also faculty in the Master of Tumor Immunology and Immunotherapy at the University Francisco de Vitoria in Madrid, Spain.

**Publications**

1. Erlotinib versus standard chemotherapy as first-line treatment for European patients with advanced EGFR mutation-positive non-small-cell lung cancer (EURTAC): a multicentre, open-label, randomised phase 3 trial. Rosell, Rafael (...) **Rolfo, Christian** (...) et al. *The Lancet Oncology*, Volume 13, Issue 3, 239 – 246
2. **Rolfo, C.**, Castiglia, M., et al., 2014. Liquid biopsies in lung cancer: The new ambrosia of researchers. *Biochimica et biophysica acta*, 1846(2), pp.539–546.
3. Exosomal Proteins in Lung Cancer: The Last Frontier in Liquid Biopsies. **Rolfo, Christian** *Journal of Thoracic Oncology*, Volume 11, Issue 10, 1609 – 1611
4. Taverna S, (...) **Rolfo C.** (...), et al. Curcumin inhibits in vitro and in vivo chronic myelogenous leukemia cells growth: a possible role for exosomal disposal of miR-21. *Oncotarget*. 2015;6(26):21918-21933.
5. A Comparison of Cell-Free DNA Isolation Kits. Sorber, Laure (...) **Rolfo, C.** (...) *The Journal of Molecular Diagnostics*, Volume 19, Issue 1, 162 – 168

## Partner 2. Prof. M<sup>a</sup> José Serrano

**Biography:** Scientific Director of Liquid Biopsy and Metastasis group. Centro GENyO. Scientific Director of oncological research based on Liquid Biopsy in the University Hospital of Granada (Oncology Unit). Visiting Professor in the University of Granada.

**Research Topics** Cancer; Liquid Biopsy, Circulating Tumor Cells (CTCs); Cell Motility; Dormancy; Exosomes; Gene Expression; Gene Regulation; Metastasis; Epigenetic; Signal Transduction; Stem Cells; miRNA

**Training Areas:** Cancer Biology

**Research:** Current Students: Diego de Miguel Lopez, Alba Rodriguez Martinez, Agustin Robles. Postdoctoral Fellows: Maria Jesus Alvarez Cubero (PhD), Clara Bayarri ( MD , PhD) Our research is based on the aim to study of the dissemination process, focused in the detection and characterization of cancer cells and exosomes and its components as DNA, RNA and microRNA using high sensitive procedures and to decode its properties using molecular biological methods. This research known as "liquid biopsy" is important for the evaluation of the prognosis of patients with lung cancer and other carcinoma, e.g. prostate, colon or breast cancer. It indicates to special targeted therapies and enables early assessment of the success or fail of therapy.

Furthermore, our more basic research investigates the process of the dissemination of tumor cells and the metastases as the main reason for malignancy of cancer. Altogether, these research activities will contribute to the development of innovative cancer therapies in the future.

Our research is largely financed by third-party funding. We are receiving funding of the Andalusian Government (CS), The Ministry of Education and Research ( FPU) and the European Union ( H2020 Program) .

## Publications

1. Francisco G. Ortega, Jose A. Lorente, Jose L. Garcia Puche , Maria P. Ruiz, Rosario M. Sanchez-Martin, Juan J. Díaz-Mochon, and Maria J. Serrano. Titulo miRNA in-situ hybridization in Circulating Tumor Cells – MishCTC. Scientific Reports (Nature) 17 March 2015.
2. Authors: Nadal R, Lorente JA, Rosell R, Serrano MJ. Título Relevance of molecular characterization of circulating tumor cells in breast cancer in the era of targeted Expert Rev Mol Diagn. 2013 Apr;13(3):295-307
3. Nadal R, Ortega FG, Salido M, Lorente JA, Rodríguez-Rivera M, Delgado-Rodríguez M, Macià M, Fernández A, Corominas JM, García Puche JL, Sánchez-Rovira P, Solé F, Serrano MJ. Título CD133 expression in circulating tumor cells from breast cancer patients: potential role in resistance to chemotherapy. Int J Cancer. 2013 Nov 15;133(10):2398-407.
4. M.J. Serrano, F.G. OrtegaMS, MJ Alvarez-Cubero, R. Nadal, P. Sanchez- Rovira , M. Salido , M. Rodríguez , J.L. García-Puche ,Delgado-Rodriguez , F. Solé , Maria A. García , Macarena Perán , , R Rosell , J.A. Marchal ,J.A. Lorente. Título Epithelial- mesenchymal transition in circulating tumor cells negative for cytokeratin expression in non-metastatic breast cancer. Oncotarget. 2014 Sep 15;5(17):7486-97.
5. Rosell R, Molina MA, Serrano MJ. EGFR mutations in circulating tumour DNA. Lancet Oncol. 2012. Oct;13(10):97

**Partner 3. Prof. Umberto Malapelle****Umberto Malapelle PhD**

Department of Public Health, University Federico II of Naples  
Via Sergio Pansini 5, 80131, Naples, Italy.

Umberto Malapelle is the Chief Supervisor of Predictive Molecular Pathology Laboratory, of the University of Naples Federico II. He was appointed in December 2015 Assistant Professor in Anatomic Pathology in School of Medicine, University of Naples Federico II.

Currently his main research interest is in the field of genomic biomarkers validation and testing for predictive information in the field of lung cancer, metastatic colorectal cancer, melanoma and gastrointestinal stromal tumor. Moreover, he has developed skills in tailoring Next Generation Assays for a number of different applications with a special focus on the simultaneous detection of clinically relevant alterations (i.e., EGFR mutations, ALK translocation, PDL – 1 expression) in the routine setting including handling of different sample types, such as tissues and/or liquid biopsy specimens.

**References:**

- 1: Malapelle U, Mayo de-Las-Casas C, Rocco D, Garzon M, Pisapia P, Jordana-Ariza N, Russo M, Sgariglia R, De Luca C, Pepe F, Martinez-Bueno A, Morales-Espinosa D, González-Cao M, Karachaliou N, Viteri Ramirez S, Bellevicine C, Molina-Vila MA, Rosell R, Troncone G. Development of a gene panel for next-generation sequencing of clinically relevant mutations in cell-free DNA from cancer patients. *Br J Cancer*. 2017.
- 2: Malapelle U, Pisapia P, Rocco D, Smeraglio R, di Spirito M, Bellevicine C, Troncone G. Next generation sequencing techniques in liquid biopsy: focus on non-small cell lung cancer patients. *Transl Lung Cancer Res*. 2016;5:505-510.
- 3: Malapelle U, Morra F, Iardi G, Visconti R, Merolla F, Cerrato A, Napolitano V, Monaco R, Guggino G, Monaco G, Staibano S, Troncone G, Celetti A. USP7 inhibitors, downregulating CCDC6, sensitize lung neuroendocrine cancer cells to PARP-inhibitor drugs. *Lung Cancer*. 2016 Jun 17. pii: S0169-30373-7.
- 4: Malapelle U, Pisapia P, Sgariglia R, Vigliar E, Biglietto M, Carlomagno C, Giuffrè G, Bellevicine C, Troncone G. Less frequently mutated genes in colorectal cancer: evidences from next-generation sequencing of 653 routine cases. *J Clin Pathol*. 2016;69:767-71.
- 5: Della Corte CM, Bellevicine C, Vicidomini G, Vitagliano D, Malapelle U, Accardo M, Fabozzi A, Fiorelli A, Fasano M, Papaccio F, Martinelli E, Troiani T, Troncone G, Santini M, Bianco R, Ciardiello F, Morgillo F. SMO Gene Amplification and Activation of the Hedgehog Pathway as Novel Mechanisms of Resistance to Anti-Epidermal Growth Factor Receptor Drugs in Human Lung Cancer. *Clin Cancer Res*. 2015;21:4686-97.

**Partner 4. Diego Cortinovis, M.D.**

Deputy Director. Oncology Unit. ASST Monza. San Gerardo Hospital. Via Pergolesi, 33 Monza, 20900 (MB) ITALY. (0039) 039-233-6040

**Education**

1990-1991 Certificate at Scientific High School "S. Allende" (Milan), with the mark of 60/60. 1994-1995 Intern at "S.Paolo" Hospital, Medicina II Division ( Director: Prof. Podda), research activity in liver disease and infirmity activity. 1995-1996 Intern at "S. Paolo", Medicina I Division Hospital (Director Prof Invernizzi), research activity in immune disease and infirmity/Day-Hospital activity.

Degree.1996-1997 Degree in Medicine and Surgery at Milan University with the mark of 110/110 "cum laude".

**Postgraduate Training**

1997-1998 Attendance to Speciality School of Chemotherapy at Milan University (Director Prof. Frascini).

2000-2003 Attendance to Speciality School of Medical Oncology at Milan University (Director Prof. Villa).

11/11/2003 Specialist in Medical Oncology

**Academic/Clinical Appointments**

1996-1998 Intern at Oncologia Medica Division, " S.Carlo" Hospital in Milan ( Director Prof. G.Luporini).

Research activity in gynaecologic, urologic, pulmonary and soft tissue tumours. Collaboration to clinical trial and to scientific paper.6/1998-12/1998 Collaboration with Palliative Care Unit "Fondazione Floriani",; service of domiciliary care for oncologic patients .1/1999-4/1999 Military service at "Scuola di Sanità e Veterinaria" in Rome, Cecchignola.4/1999-3/2000 Medical Lieutenant at "Caserma S.Barbara" Milan with occasional employment as Sanitary Service Manager.4/2000-11/2003 Postgraduate at "S.C. Oncologia Medica B", Istituto Nazionale per lo Studio e la Cura dei Tumori, Milan (Director Prof. E.Bajetta).

2/2004-08/2006 Fellow at " S.C. Oncologia Medica B" Istituto Nazionale per lo Studio e la cura dei Tumori, Milan (Director Prof. E. Bajetta).08/2006 -02/2008 Solid tumor Oncologist of the Operative Unit 2 of Medical Oncology at Fondazione IRCCS Istituto Nazionale dei Tumori (Director Prof. E. Bajetta)

**02/2008 – 11/2015** Solid tumor Oncologist of the Medical Oncology Unit at A.S.S.T. Monza (ex S. Gerardo Hospital) – Monza (Director Prof. P.Bidoli)

**12/2015 to present** Deputy Director of the Medical Oncology (specialized in thoracic and lung cancer) Unit at A.S.S.T. Monza, (Director Prof. P.Bidoli).

**Main domain of research**

Lung, mesothelioma, with a strong focus in clinical trials of new diagnostic and treatment approaches and develop methods for improved treatment of cancer.

**Most recent peer reviewed publications**

1. First-line ceritinib versus platinum-based chemotherapy in advanced ALK-rearranged non-small-cell lung cancer (ASCEND-4): a randomised, open-label, phase 3 study. Soria JC, (...) Cortinovis D, (...)de Castro G Jr. Lancet. 2017 Jan 23. pii: S0140-6736(17)30123-X. doi: 10.1016/S0140-6736(17)30123-X. [Epub ahead of print]

2. Atezolizumab versus docetaxel in patients with previously treated non-small-cell lung cancer (OAK): a phase 3, open-label, multicentre randomised controlled trial. Rittmeyer A, (...), Cortinovis DL, (...), Gandara DR; OAK Study Group..

3. Focus on Nivolumab in NSCLC. Cortinovis DL, Canova S, Abbate M, Colonese F, Bidoli P. Front Med (Lausanne). 2016 Dec 13;3:67. doi: 10.3389/fmed.2016.00067.

4.Efficacy and safety of rechallenge treatment with gefitinib in patients with advanced non-small cell lung cancer. Cappuzzo F, (...) Cortinovis D, (...),De Marinis F. Lung Cancer. 2016 Sep;99:31-7. doi: 10.1016/j.lungcan.2016.06.008.

5.BE-POSITIVE: Beyond progression after tyrosine kinase inhibitor in EGFR- positive non small cell lung cancer patients: Results from a multicenter Italian observational study. Vavalà T, (...), Cortinovis D, (...) Novello S. Lung Cancer. 2016 May;95:73-81. doi: 10.1016/j.lungcan.2016.02.011.

**Partner 5. Prof. James Yang**

Dr James Chih-Hsin Yang is Professor at the Graduate Institute of Oncology and joint appointed in the Graduate Institute of Clinical Medicine and the Graduate Institute of Clinical Pharmacy at the College of Medicine, National Taiwan University (NTU). He has been a staff member in the Department of Oncology at the University Hospital since 1995. Dr Yang received his MD in 1986 and completed his internal medicine residency at the National Taiwan University Hospital. Between 1992 and 1995 he undertook oncology fellowship training at the National Cancer Institute at Bethesda, Maryland. During that period, he was involved with many NCI phase I studies and NCI sponsored-protocol development for all phases of cancer clinical trials. He completed his PhD degree between 1996 and 2000 at the Graduate Institute of Clinical Medicine, NTU. Currently he serves as deputy director of Department of Oncology at NTU Hospital.

Dr Yang's research focuses on lung cancer treatment and the mechanism of multidrug resistance of chemotherapy or targeted therapy. His basic research works included molecular mechanisms of resistance and reversal of resistance to chemotherapeutic agents and tyrosine kinase inhibitors. Dr. Yang currently served as the executive secretary of Taiwan Lung Cancer Clinical Trial Consortium (TALCC). He has authored more than 100 papers published in international peer-reviewed journals in the field of lung cancer research. Currently he serves in the editorial board of Journal of Clinical Oncology, the associate editor of Journal of Thoracic Oncology, Asia Pacific Journal of Clinical Oncology and Journal of the Formosan Medical Association. Dr. Yang is the member of American Society of Clinical Oncology, American Association for Cancer Research, European Society of Medical Oncology and International Association for the Study of Lung Cancer. Dr. Yang is a well-recognized world key opinion leader in lung cancer clinical studies, especially in the development of irreversible EGFR tyrosine kinase inhibitors. He is an investigator and served as the principle investigator for many international lung cancer clinical trials, such as afatinib pivotal studies. He has delivered more than 100 invited speech in international symposium and conferences. He attended many advisory boards for clinical trial preparation and drug development. He recently received 2nd Kobayashi Foundation Cancer Research Award during Asian Clinical Oncology Society (ACOS) meeting in 2012.

**References**

1. Wu, Y.-L., Sequist, L. V., Hu, C.-P., Feng, J., Lu, S., Huang, Y., ... Yang, J. C.-H. (2017). EGFR mutation detection in circulating cell-free DNA of lung adenocarcinoma patients: analysis of LUX-Lung 3 and 6. *British Journal of Cancer*, 116(2), 175–185. <http://doi.org/10.1038/bjc.2016.420>
2. Lin, K.-H., Hong, S.-T., Wang, H.-T., Lo, Y.-L., Lin, A. M.-Y., & Yang, J. C.-H. (2016). Enhancing Anticancer Effect of Gefitinib across the Blood–Brain Barrier Model Using Liposomes Modified with One  $\alpha$ -Helical Cell-Penetrating Peptide or Glutathione and Tween 80. *International Journal of Molecular Sciences*, 17(12), 1998. <http://doi.org/10.3390/ijms17121998>
3. Genetic Modifiers of Progression-free Survival in Never-smoking Lung Adenocarcinoma Patients Treated with First-line TKIs. Chang, Dr. I-Shou; Jiang, Dr. Shih Sheng; Yang, Dr. James Chih-Hsin; Su, Dr. Wu-Chou; Chien, Dr. Li-Hsin et al. (2016). *American Journal of Respiratory And Critical Care Medicine*
4. Profile of the theascreen® EGFR RGQ PCR kit as a companion diagnostic for gefitinib in non-small cell lung cancer. Emily Han-Chung Hsiue, Jih-Hsiang Lee, Chia-Chi Lin, and James Chih-Hsin Yang. *Expert Review Of Molecular Diagnostics* Vol. 16 , Iss. 12, 2016
5. Association Between Plasma Genotyping and Outcomes of Treatment With Osimertinib (AZD9291) in Advanced Non–Small-Cell Lung Cancer. Geoffrey R. (...), and James Chih-Hsin Yang, National Taiwan University and National Taiwan University Hospital, Taipei, Taiwan. *Journal of Clinical Oncology* 2016 34:28, 3375-3382

**12. Global financial plan: sum of year 1-3. Please describe the requested budget only.****(Please note that eligibility of costs is subject to national rules and regulations: refer to Annex 4 of the Call Text).**

|                                       |                     |                             |                    |                    |                                    |           |           |
|---------------------------------------|---------------------|-----------------------------|--------------------|--------------------|------------------------------------|-----------|-----------|
| Acronym:                              |                     |                             |                    |                    |                                    |           |           |
| No.                                   | Project coordinator | Partner 2                   | Partner 3          | Partner 4          | Partner 5                          | Partner 6 | Partner 7 |
| Name (principal investigator)         | Christian Rolfo     | M <sup>a</sup> José Serrano | Umberto Malapelle  | Diego Cortinovis   | James Chih-Hsin Yang               |           |           |
| Country                               | BELGIUM             | SPAIN                       | ITALY              | ITALY              | Taiwan(ROC)                        |           |           |
| Funding organisation                  | FWO                 | AECC/ISCIII                 | MINISTRY OF HEALTH | MINISTRY OF HEALTH | Ministry of Science and Technology |           |           |
| Personnel (€)                         |                     |                             |                    |                    |                                    |           |           |
| - Scientist                           |                     |                             |                    |                    |                                    |           |           |
| - PhD-Student                         |                     | 88,500                      |                    |                    | 0                                  |           |           |
| - Technician                          |                     |                             |                    | 20,000             | 0                                  |           |           |
| - Other                               | 15,000              |                             |                    | 33,000             | 60,000                             |           |           |
| Person months                         |                     |                             |                    |                    | 0                                  |           |           |
| - Scientist                           |                     |                             |                    |                    |                                    |           |           |
| - PhD-Student                         |                     | 24                          |                    |                    | 0                                  |           |           |
| - Technician                          |                     |                             |                    |                    | 0                                  |           |           |
| - Other                               | 12                  |                             |                    |                    | 36                                 |           |           |
| Consumables (€)                       | 182,404             | 79,000                      | 18,600             | 1,000              | 330,000                            |           |           |
| Equipment (€)                         |                     |                             |                    |                    | 100,000                            |           |           |
| Study/Clinical trial (€) <sup>1</sup> |                     |                             |                    |                    | 0                                  |           |           |
| Travel (€) <sup>2</sup>               | 4,000               | 3,000                       | 3,000              | 3,000              | 0                                  |           |           |
| Capacity building (€) <sup>3</sup>    |                     |                             |                    |                    | 0                                  |           |           |
| Other direct costs (€) <sup>4</sup>   | 2,000               | 3,000                       |                    | 3,000              | 110,000                            |           |           |
| (national) Overheads (€)              |                     |                             |                    |                    |                                    |           |           |
| <b>Total requested budget (€)</b>     | 199,404             | 190,720                     | 21,600             | 83,000             | 500,000                            |           |           |

<sup>1</sup> If applicable: incl. clinical trial drugs/compounds, clinical trial fees and insurance.<sup>2</sup>Travel expenses should include the participation of the coordinators and/or principal investigators in an intermediate and/or a final status symposium to present the results of their projects (organized by the Joint Call Secretariat).



<sup>3</sup> Separate budget for capacity building activities (if eligible for the funding organization/country).

<sup>4</sup> e.g. subcontracting, provisions, licensing fees.

### 13. Individual financial plans: sum of year 1-3.

**(Please note that eligibility of costs is subject to national/regional rules and regulations: refer to Annex 4 of the call)**

#### 13.1

|                          |                                                                                 |                                                                                                                                                                                                                                         |
|--------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Partner name:            | University Hospital of Antwerp – Phase I Unit / Centre for Oncological Research |                                                                                                                                                                                                                                         |
| Funding organisation     | FWO                                                                             |                                                                                                                                                                                                                                         |
| Country                  | BELGIUM                                                                         |                                                                                                                                                                                                                                         |
|                          | Requested budget                                                                | Justification                                                                                                                                                                                                                           |
| Personnel (€)            | 15,000                                                                          | <i>Part-time Data Manager</i>                                                                                                                                                                                                           |
| Consumables (€)          | 178,404                                                                         | <i>Kits for RNA extraction from Exosomes – 1 Kit per 50 patients - Cost for 1 kit €1202 - Total Cost (100 patients) = €2,404. Panel for Exosomal Genes identificacion – Cost €176 per sample = €176,000. Disposable material – 500€</i> |
| Equipment (€)            | 0                                                                               | <i>N/A</i>                                                                                                                                                                                                                              |
| Study/Clinical trial (€) |                                                                                 | <i>N/A</i>                                                                                                                                                                                                                              |
| Travel (€)               | 4,000                                                                           | <i>Two consortium meetings per year, total 6 meetings at an average cost of €500 per travel and accommodation</i>                                                                                                                       |
| Capacity building (€)    | 0                                                                               | <i>N/A</i>                                                                                                                                                                                                                              |
| Other direct costs (€)   | 2,000                                                                           | <i>Courier charges related to sample delivery from hospital to laboratories. Microscopy and genomic services charges. Data base.</i>                                                                                                    |
| Overheads (€)            | 0                                                                               | <i>N/A</i>                                                                                                                                                                                                                              |
| <b>Total budget (€)</b>  | <b>199,404</b>                                                                  |                                                                                                                                                                                                                                         |

## 13.2

|                          |                                                            |                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Partner name:            | Complejo Hospitalario de Granada – Fundación FIBAO - GENyO |                                                                                                                                                                                                                                                                                                                                                                                        |
| Funding organisation     | AECC                                                       |                                                                                                                                                                                                                                                                                                                                                                                        |
| Country                  | SPAIN                                                      |                                                                                                                                                                                                                                                                                                                                                                                        |
|                          | Requested budget                                           | Justification                                                                                                                                                                                                                                                                                                                                                                          |
| Personnel (€)            | 88,500                                                     | <i>One PM during 2 years, with a Master in science degree (biology specialization). This person will be responsible of the following tasks: performing, plasma isolation, isolation of CTC and cfDNA, molecular and cellular analysis including, CTC characterization, immunohistochemistry, PCR and NGS. QW12 PMs with a Master in Science degree (bioinformatics specialization)</i> |
| Consumables (€)          |                                                            |                                                                                                                                                                                                                                                                                                                                                                                        |
| Equipment (€)            |                                                            |                                                                                                                                                                                                                                                                                                                                                                                        |
| Study/Clinical trial (€) |                                                            |                                                                                                                                                                                                                                                                                                                                                                                        |
| Travel (€)               | 3,000                                                      | <i>Two consortium meetings per year, total 6 meetings at an average cost of €500 per travel and accommodation</i>                                                                                                                                                                                                                                                                      |
| Capacity building (€)    |                                                            |                                                                                                                                                                                                                                                                                                                                                                                        |
| Other direct costs (€)   |                                                            |                                                                                                                                                                                                                                                                                                                                                                                        |
| Overheads (€)            |                                                            |                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Total budget (€)</b>  | <b>91,500</b>                                              |                                                                                                                                                                                                                                                                                                                                                                                        |

|                          |                                                            |                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Partner name:            | Complejo Hospitalario de Granada – Fundación FIBAO - GENyO |                                                                                                                                                                                                                                                                                                                                                 |
| Funding organisation     | ISCIII                                                     |                                                                                                                                                                                                                                                                                                                                                 |
| Country                  | SPAIN                                                      |                                                                                                                                                                                                                                                                                                                                                 |
|                          | Requested budget                                           | Justification                                                                                                                                                                                                                                                                                                                                   |
| Personnel (€)            |                                                            |                                                                                                                                                                                                                                                                                                                                                 |
| Consumables (€)          | 79,000                                                     | <i>Kits Isoflux for isolating and detection of CTCs (1 kit per 8 patients – Cost of 1 kit €1,500 – total cost (100 patients): €18,750; NGS for gene panel: €150 per sample – total cost: €15,000; Ficoll €600 per 10 patients- total cost: €6,000; Blood tubes: €1,500; primary and labelled antibodies: €12,000; disposable material: €985</i> |
| Equipment (€)            |                                                            |                                                                                                                                                                                                                                                                                                                                                 |
| Study/Clinical trial (€) |                                                            |                                                                                                                                                                                                                                                                                                                                                 |
| Travel (€)               |                                                            |                                                                                                                                                                                                                                                                                                                                                 |
| Capacity building (€)    |                                                            |                                                                                                                                                                                                                                                                                                                                                 |
| Other direct costs (€)   | 3,000                                                      | <i>Courier charges related to sample delivery from hospital to laboratories. Microscopy and genomic services charges</i>                                                                                                                                                                                                                        |
| Overheads (€)            | 17,220                                                     | <i>21% of ISCIII eligible direct costs</i>                                                                                                                                                                                                                                                                                                      |
| <b>Total budget (€)</b>  | <b>99,220</b>                                              |                                                                                                                                                                                                                                                                                                                                                 |

## 13.3

|                          |                                                                |                                                                                                            |
|--------------------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Partner name:            | Departament of Public health- University Federico II of Naples |                                                                                                            |
| Funding organisation     | MINISTRY OF HEALTH                                             |                                                                                                            |
| Country                  | ITALY                                                          |                                                                                                            |
|                          | Requested budget                                               | Justification                                                                                              |
| Personnel (€)            | 0                                                              | N/A                                                                                                        |
| Consumables (€)          | 18,600                                                         | DL8 IonChef HIQ Kit, IonChef Sequencing 200 Kit and 316 PGM chip v 2 (ThermoFisher)                        |
| Equipment (€)            | 0                                                              | N/A                                                                                                        |
| Study/Clinical trial (€) |                                                                | N/A                                                                                                        |
| Travel (€)               | 3,000                                                          | Two consortium meetings per year, total 6 meetings at an average cost of €500 per travel and accommodation |
| Capacity building (€)    | 0                                                              | N/A                                                                                                        |
| Other direct costs (€)   | 0                                                              | N/A                                                                                                        |
| Overheads (€)            | 0                                                              | N/A                                                                                                        |
| <b>Total budget (€)</b>  | 21.600                                                         |                                                                                                            |

## 13.4

|                          |                                                  |                                                                                                                                                                      |
|--------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Partner name:            | San Gerardo Hospital. Oncology Unit. ASST Monza. |                                                                                                                                                                      |
| Funding organisation     | MINISTRY OF HEALTH                               |                                                                                                                                                                      |
| Country                  | ITALY                                            |                                                                                                                                                                      |
|                          | Requested budget                                 | Justification                                                                                                                                                        |
| Personnel (€)            | 53,000                                           | 1 part-time dedicated technician: €20,000. 1 data manager €33,000                                                                                                    |
| Consumables (€)          | 4,000                                            | Disposable materials (Pipettes, needles, tubes...)                                                                                                                   |
| Equipment (€)            | 0                                                | N/A                                                                                                                                                                  |
| Study/Clinical trial (€) |                                                  | N/A                                                                                                                                                                  |
| Travel (€)               | 3,000                                            | Two consortium meetings per year, total 6 meetings at an average cost of €500 per travel and accommodation                                                           |
| Capacity building (€)    | 0                                                | N/A                                                                                                                                                                  |
| Other direct costs (€)   | 23,000                                           | Courier charges related to sample delivery from hospital to laboratories. Microscopy and genomic services charges. Secretary services, ethical committee procedures. |
| Overheads (€)            |                                                  | Please refer to your national regulations before calculating overheads                                                                                               |
| <b>Total budget (€)</b>  | <b>83,000</b>                                    |                                                                                                                                                                      |

## 13.5

|                          |                                    |                                                                                                                                                                                                                                       |
|--------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Partner name:            | James Chih-Hsin Yang               |                                                                                                                                                                                                                                       |
| Funding organisation     | Ministry of Science and Technology |                                                                                                                                                                                                                                       |
| Country                  | Taiwan                             |                                                                                                                                                                                                                                       |
|                          | Requested budget                   | Justification                                                                                                                                                                                                                         |
| Personnel (€)            | 60,000                             | A study coordinate will be hired for the project. He/she will screen potential subjects, collect samples, manage project and manage other administration issues. 20,000€ per year is required to cover his/her salary and insurances. |
| Consumables (€)          | 230,000                            | 1. Deep sequencing of target genes (230,000€)                                                                                                                                                                                         |
| Equipment (€)            | 100,000                            | 1. Refrigerators for storage of samples (10,000€)<br>2. Maintenance fee for the next generation sequencers (30,000€ per year = 90,000€ in total)                                                                                      |
| Study/Clinical trial (€) | 0                                  |                                                                                                                                                                                                                                       |
| Travel (€)               | 0                                  |                                                                                                                                                                                                                                       |
| Capacity building (€)    | 0                                  |                                                                                                                                                                                                                                       |
| Other direct costs (€)   | 110,000                            | 1. Approval from IRB and regulatory authorities (3,000€)<br>2. Miscellaneous issues (7,000 €)<br>3. Preparation and international shipping of samples (100,000€)                                                                      |
| Overheads (€)            |                                    |                                                                                                                                                                                                                                       |
| <b>Total budget (€)</b>  | <b>500,000</b>                     |                                                                                                                                                                                                                                       |

## 13.6

|                          |                  |                                                                                                                                                                                                                   |
|--------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Partner name:            |                  |                                                                                                                                                                                                                   |
| Funding organisation     |                  |                                                                                                                                                                                                                   |
| Country                  |                  |                                                                                                                                                                                                                   |
|                          | Requested budget | Justification                                                                                                                                                                                                     |
| Personnel (€)            |                  | <i>Please indicate the number of PMs per type of personnel, indicating the project tasks that justify the inclusion of that number of PMs</i>                                                                     |
| Consumables (€)          |                  | <i>Please identify the consumables to be included, and their importance within your projects' tasks and objectives</i>                                                                                            |
| Equipment (€)            |                  | <i>Please indicate and justify the equipment to be acquired in accordance to project tasks and objectives. Applicants should also check if equipment is eligible in accordance to their national regulations.</i> |
| Study/Clinical trial (€) |                  | <i>Please indicate the concrete participation/work package(s) in the study/clinical trial</i>                                                                                                                     |
| Travel (€)               |                  | <i>Please give an estimate on the number and main reasons for the travels within the project</i>                                                                                                                  |
| Capacity building (€)    |                  | <i>Please indicate the type of capacity building and necessary efforts (PMs, travel etc.).</i>                                                                                                                    |
| Other direct costs (€)   |                  | <i>May include subcontracting, fees, insurances, etc. Please justify each predicted expenditure with relation to project tasks and objectives</i>                                                                 |
| Overheads (€)            |                  | <i>Please refer to your national regulations before calculating overheads</i>                                                                                                                                     |
| <b>Total budget (€)</b>  |                  |                                                                                                                                                                                                                   |

## 13.7

|                          |                  |                                                                                                                                                                                                                   |
|--------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Partner name:            |                  |                                                                                                                                                                                                                   |
| Funding organisation     |                  |                                                                                                                                                                                                                   |
| Country                  |                  |                                                                                                                                                                                                                   |
|                          | Requested budget | Justification                                                                                                                                                                                                     |
| Personnel (€)            |                  | <i>Please indicate the number of PMs per type of personnel, indicating the project tasks that justify the inclusion of that number of PMs</i>                                                                     |
| Consumables (€)          |                  | <i>Please identify the consumables to be included, and their importance within your projects' tasks and objectives</i>                                                                                            |
| Equipment (€)            |                  | <i>Please indicate and justify the equipment to be acquired in accordance to project tasks and objectives. Applicants should also check if equipment is eligible in accordance to their national regulations.</i> |
| Study/Clinical trial (€) |                  | <i>Please indicate the concrete participation/work package(s) in the study/clinical trial</i>                                                                                                                     |
| Travel (€)               |                  | <i>Please give an estimate on the number and main reasons for the travels within the project</i>                                                                                                                  |
| Capacity building (€)    |                  | <i>Please indicate the type of capacity building and necessary efforts (PMs, travel etc.).</i>                                                                                                                    |
| Other direct costs (€)   |                  | <i>May include subcontracting, fees, insurances, etc. Please justify each predicted expenditure with relation to project tasks and objectives</i>                                                                 |
| Overheads (€)            |                  | <i>Please refer to your national regulations before calculating overheads</i>                                                                                                                                     |
| <b>Total budget (€)</b>  |                  |                                                                                                                                                                                                                   |



#### 14. Reviewers

Please note that providing the information below is optional. The Call Steering Committee (CSC) will consider these suggestions provided that they do not interfere with the objective and thorough evaluation of the proposal.

**Suggested reviewers for reviewing this proposal (up to five), without a conflict of interest (i.e. not working in the same institute, no co-publication in the past 5 years).**

| Name | Institute | Email address |
|------|-----------|---------------|
|      |           |               |
|      |           |               |
|      |           |               |
|      |           |               |
|      |           |               |

**Reviewers to be excluded from reviewing this proposal (up to five)**

| Name | Institute | Email address |
|------|-----------|---------------|
|      |           |               |
|      |           |               |
|      |           |               |
|      |           |               |
|      |           |               |

#### 15. IF APPLICABLE:

A signed written confirmation that the project partner with own funding (also from a country/region not participating in the JTC 2016) has secured his/her funding.

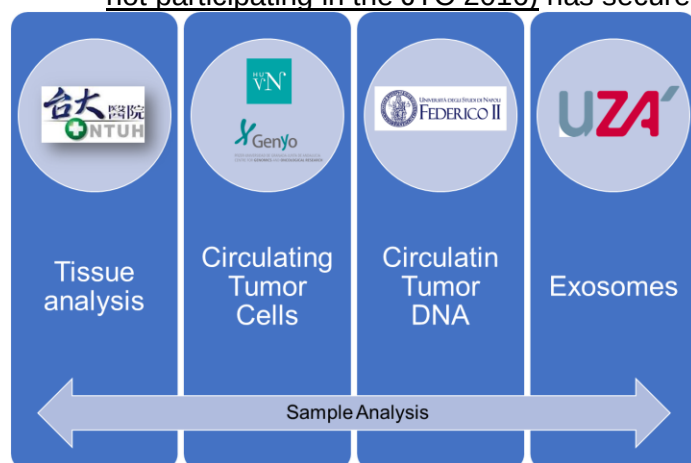
#### **PLEASE NOTE**

- **Proposals that do not meet the national eligibility criteria and requirements will be declined without further review.**
- **Proposals must be sent in one single PDF document.**
- **Proposals exceeding the length limitations (of each section) will be discarded without further review:**

**Cut-off (incl. all appendices): 30 pages**

**Annexes**(to be included in this document)

- Diagrams and figures
- A signed written confirmation that the project partner with own funding (also from a country/region not participating in the JTC 2016) has secured his/her funding.

**Fig.1. Responsible of each area of the analytical part of the project.**

| LIQUIDLUNG                                                                         | Months |   |   |    |    |    |    |    |    |    |    |    |
|------------------------------------------------------------------------------------|--------|---|---|----|----|----|----|----|----|----|----|----|
| Tasks                                                                              | 3      | 6 | 9 | 12 | 15 | 18 | 21 | 24 | 27 | 30 | 33 | 36 |
| <b>WP 1. Project Management (ALL)</b>                                              |        |   |   |    |    |    |    |    |    |    |    |    |
| T 1.1. Administrative coordination                                                 |        |   |   |    |    |    |    |    |    |    |    |    |
| T 1.2. Control project progress                                                    |        |   |   |    |    |    |    |    |    |    |    |    |
| T 1.3. Control quality standards                                                   |        |   |   |    |    |    |    |    |    |    |    |    |
| T 1.4. Control EU Legal obligations                                                |        |   |   |    |    |    |    |    |    |    |    |    |
| <b>WP 2. Patient recruitment and sample collection (ALL)</b>                       |        |   |   |    |    |    |    |    |    |    |    |    |
| T 2.1. Patient selection                                                           |        |   |   |    |    |    |    |    |    |    |    |    |
| T 2.2. Informed consent signatures from patients                                   |        |   |   |    |    |    |    |    |    |    |    |    |
| T 2.3. Patient accrual and clinical data form compilation                          |        |   |   |    |    |    |    |    |    |    |    |    |
| T 2.4 Sample extraction and preparation                                            |        |   |   |    |    |    |    |    |    |    |    |    |
| <b>WP 3. Molecular analysis (GRANADA – ITALY – BELGIUM-TAIWAN)</b>                 |        |   |   |    |    |    |    |    |    |    |    |    |
| T 3.1. Molecular characterization through NGS of Circulating Tumor Cells (GRANADA) |        |   |   |    |    |    |    |    |    |    |    |    |
| T 3.2. Molecular characterization of cell-free DNA (ITALY)                         |        |   |   |    |    |    |    |    |    |    |    |    |
| T 3.3. Molecular characterization of exosomes from plasma (BELGIUM)                |        |   |   |    |    |    |    |    |    |    |    |    |
| T 3.4. Molecular characterization of tissue (TAIWAN)                               |        |   |   |    |    |    |    |    |    |    |    |    |
| <b>WP 4. Data Analysis (GRANADA)</b>                                               |        |   |   |    |    |    |    |    |    |    |    |    |
| T 4.1. Statistical analysis of the results                                         |        |   |   |    |    |    |    |    |    |    |    |    |

**Fig.2. Flowchart of the WorkPackages for the 36 months of the project.**